Lab Assignment 02

BRAC UNIVERSITY

Inspiring Excellence

Course Code:	**CSE111**
Course Title:	**Programming Language II**
Topic:	**OOP Basics, Instance Variable and Instance Method**
Number of Tasks:	**12 (Classwork: 06, Homework: 06)**

[Submit all the Coding Tasks (Homework: Task 1 to 4) in the Google Form shared on buX before the next lab. Submit the Tracing Tasks (Homework: Task 5 to 6) handwritten to your Lab Instructors at the beginning of the lab]

[You are not allowed to change the driver codes of any of the tasks]

CLASSWORK

Task 1

You are given the following "**University**" class:

```
public class University{
    public String name;
    public String country;
}
```

Now write a Java **tester** class named "**UniversityTester**".

a. Write the main method and create 2 objects of **University** class and print the location of the objects and print the instance variables of the objects. Are the location of the objects the same?

b. Now change the instance variables of the first object.
   name = "Imperial College London"
   country = "England"

   Now change the instance variables of the second object.
   name = "BRAC University"
   country = "Bangladesh"

   Now check if the instance variables of both objects have changed or not and whether the instance variables of both objects are of the same value or not.

Task 2

Complete the **“Cat”** class so the main method produces the following output:

<table>
<tr><th>Test Class</th><th>Output</th></tr>
<tr><td><pre>
public class Test7{
    public static void main(String [] args){
        Cat c1 = new Cat();
        System.out.println("1==================");
        c1.printCat();
        c1.color = "Black";
        System.out.println("2==================");
        c1.printCat();
        c1.color = "Brown";
        c1.action = "jumping";
        System.out.println("3==================");
        c1.printCat();
    }
}
</pre></td><td><pre>
1==================
White cat is sitting
2==================
Black cat is sitting
3==================
Brown cat is jumping
</pre></td></tr>
</table>

Task 3

Design the **Course** class to generate the correct output from the driver code provided below:

<table>
<tr><th>Driver Code</th><th>Output</th></tr>
<tr><td><pre>
public class Tester3{
  public static void main(String[] args) {
    Course c1 = new Course();
    Course c2 = new Course();
    c1.updateDetails("Programming Language I","CSE110", 3);
    System.out.println("========== 1 ==========");
    c1.displayCourse();
    c2.updateDetails("Data Structures","CSE220", 3);
    System.out.println("========== 2 ==========");
    c2.displayCourse();
    c1.updateDetails("Programming Language II","CSE111", 3);
    System.out.println("========== 3 ==========");
    c1.displayCourse();
  }
}
</pre></td><td><pre>
========== 1 ==========
Course Name:
Programming Language I
Course Code: CSE110
Course Credit: 3
========== 2 ==========
Course Name: Data
Structures
Course Code: CSE220
Course Credit: 3
========== 3 ==========
Course Name:
Programming Language II
Course Code: CSE111
Course Credit: 3
</pre></td></tr>
</table>

Task 4

Design the **CellPhone** class so that the **main** method of tester class can produce the following output:

<table>
<tr><th>Tester Class</th><th>Output</th></tr>
<tr><td>

```java
public class Tester4{
  public static void main(String[]args){
     CellPhone phone1 = new CellPhone();
     phone1.printDetails();
     phone1.model ="Nokia 1100";
     System.out.println("1##################");
     phone1.storeContact("Joy - 01834");
     System.out.println("===================");
     phone1.printDetails();
     System.out.println("2##################");
     phone1.storeContact("Toya - 01334");
     phone1.storeContact("Aayan - 01135");
     System.out.println("===================");
     phone1.printDetails();
     System.out.println("3##################");
     phone1.storeContact("Sani - 01441");
     System.out.println("===================");
     phone1.printDetails();
  }
}
```

</td><td>

```
Phone Model unknown
Contacts Stored 0
1##################
Contact Stored
===================
Phone Model Nokia 1100
Contacts Stored 1
Stored Contacts:
Joy - 01834
2##################
Contact Stored
Contact Stored
===================
Phone Model Nokia 1100
Contacts Stored 3
Stored Contacts:
Joy - 01834
Toya - 01334
Aayan - 01135
3##################
Memory full. New contact can't be
stored.
===================
Phone Model Nokia 1100
Contacts Stored 3
Stored Contacts:
Joy - 01834
Toya - 01334
Aayan - 01135
```

</td></tr>
</table>

Task 5

```java
public class Task5 {
    public int p = 3, y = 2, sum;
    public void methodA(){
        int x = 0, y = 0;
        y = y + this.y;
        x = sum + 2 + p;
        sum = x + methodB(p, y) + y;
        System.out.println(x + " " + y+ " " + sum);
    }
    public int methodB(int p, int n){
        int x = 0;
        y = y + (++p);
        x = x + 2 + n;
        sum = sum + x + y;
        System.out.println(x + " " + y+ " " + sum);
        return sum;
    }
}
```

Driver code:

```java
public class Tester5 {
    public static void main(String [] args){
        Task5 t1 = new Task5();
        t1.methodA();
        t1.methodA();
    }
}
```

Outputs		
x	y	Sum

Task 6

```java
public class Test6{
  public int sum;
  public int y;
  public void methodA(){
    int x=2, y =3;
    int [] msg ={3, 7};
    y = this.y + msg[0];
    methodB(msg[1]++, msg[0]);
    x = x + this.y + msg[1];
    sum = x + y + msg[0];
    System.out.println(x + " " + y+ " " + sum);
  }
  public void methodB(int mg2, int mg1){
    int x = 0;
    y = this.y + mg2;
    x = x + 19 + mg1;
    sum = this.sum + x + y;
    mg2 = y + mg1;
    mg1 = mg2 + x + 2;
    System.out.println(x + " " + y+ " " + sum);
  }
}
```

Driver code:

```java
public class Tester6{
  public static void main (String args[]){
    Test6 t1 = new Test6();
    t1.methodB(5,-8);
    Test6 t2 = new Test6();
    t2.methodA();
  }
}
```

Outputs		

HOMEWORK

Task 1

Design the "**ImaginaryNumber**" class to generate the **output** given below:

<table>
<tr><th>Tester Class</th><th>Output</th></tr>
<tr><td><pre>public class Tester7{
  public static void main(String [] args){
     ImaginaryNumber num1 = new ImaginaryNumber();
     String p = num1.printNumber();
     System.out.println(p);
     System.out.println("1********");
     num1.realPart=3;
     num1.imaginaryPart=7;
     System.out.println(num1.printNumber());
     System.out.println("2********");
     ImaginaryNumber num2 = new ImaginaryNumber();
     num2.realPart=1;
     num2.imaginaryPart=9;
     System.out.println(num2.printNumber());
    }
}</pre></td><td><pre>0 + 0i
1********
3 + 7i
2********
1 + 9i</pre></td></tr>
</table>

Task 2

Implement the "**Assignment**" class with necessary properties, so that the given output is produced for the provided driver code.

<table>
<tr><th>Driver Class</th><th>Output</th></tr>
<tr><td><pre>public class AssignmentTester{
  public static void main(String [] args){
    Assignment as1 = new Assignment();
    as1.printDetails();
    System.out.println("1---------------");
    as1.tasks = 11;
    as1.difficulty = "Moderate";
    as1.submission = true;
    as1.printDetails();</pre></td><td><pre>Number of tasks: 0
Difficulty level: null
Submission required: false
1---------------
Number of tasks: 11
Difficulty level: Moderate
Submission required: true
2---------------
Assignment will not require</pre></td></tr>
</table>

<pre>   System.out.println("2---------------");
    System.out.println(as1.makeOptional());
    System.out.println("3---------------");
    as1.printDetails();
    System.out.println("4---------------");
    Assignment as2 = new Assignment();
    as2.tasks = 12;
    as2.difficulty = "Hard";
    as2.submission = false;
    as2.printDetails();
    System.out.println("5---------------");
    System.out.println(as2.makeOptional());
  }
}</pre>
	<pre>submission
	3---------------
	Number of tasks: 11
	Difficulty level: Moderate
	Submission required: false
	4---------------
	Number of tasks: 12
	Difficulty level: Hard
	Submission required: false
	5---------------
	Submission is already not required</pre>

Task 3

Create an **Employee** class to provide the expected output.

- An employee will have a name, salary and designation.
- The name will be assigned inside the newEmployee() method
- Whenever a New Employee joins his/her salary will be **Tk. 30,000** and the designation will be **junior**.
- Employees with salaries greater than **Tk. 50,000** and **Tk. 30,000** need to pay **30%** and **10%** of salary as tax respectively.
- Employees can be promoted to **senior**, **lead** and **manager** positions. Based on their promotion they will get an increment of **Tk. 25,000**, **Tk. 50,000** and **Tk. 75,000** respectively.

<table>
<tr><th>Driver Code</th><th>Expected Output</th></tr>
<tr><td>

```java
public class Tester9{
  public static void main(String[] args){

    Employee emp1 = new Employee();
    Employee emp2 = new Employee();
    Employee emp3 = new Employee();

    emp1.newEmployee("Harry Potter");
    emp2.newEmployee("Hermione Granger");
    emp3.newEmployee("Ron Weasley");
    System.out.println("1 ==========");
    emp1.displayInfo();
    System.out.println("2 ==========");
    emp2.displayInfo();
    System.out.println("3 ==========");
    emp3.displayInfo();
    System.out.println("4 ==========");
    emp1.calculateTax();
    System.out.println("5 ==========");
    emp1.promoteEmployee("lead");
    System.out.println("6 ==========");
    emp1.calculateTax();
    System.out.println("7 ==========");
    emp1.displayInfo();
    System.out.println("8 ==========");
    emp3.promoteEmployee("manager");
    System.out.println("9 ==========");
    emp3.calculateTax();
    System.out.println("10 ==========");
    emp3.displayInfo();
  }
}
```

</td><td>

```
1 ==========
Employee Name: Harry Potter
Employee Salary: 30000.0 Tk
Employee Designation: junior
2 ==========
Employee Name: Hermione Granger
Employee Salary: 30000.0 Tk
Employee Designation: junior
3 ==========
Employee Name: Ron Weasley
Employee Salary: 30000.0 Tk
Employee Designation: junior
4 ==========
No need to pay tax
5 ==========
Harry Potter has been promoted to lead
New Salary: 80000.00 Tk
6 ==========
Harry Potter Tax Amount: 24000.0 Tk
7 ==========
Employee Name: Harry Potter
Employee Salary: 80000.0 Tk
Employee Designation: lead
8 ==========
Ron Weasley has been promoted to manager
New Salary: 105000.00 Tk
9 ==========
Ron Weasley Tax Amount: 31500.0 Tk
10 ==========
Employee Name: Ron Weasley
Employee Salary: 105000.0 Tk
Employee Designation: manager
```

</td></tr>
</table>

Task 4

Implement the “**MobilePhone**” class with necessary properties to produce the given output for the provided driver code.

<table>
<tr><th>Driver Class</th><th>Output</th></tr>
<tr><td>

```
public class MobilePhoneTester{
  public static void main(String args []){
    MobilePhone m1 = new MobilePhone();
    MobilePhone m2 = new MobilePhone();
    m1.setContactCapacity(5);
    m2.setContactCapacity(100);
    m1.details();
    System.out.println("1----------------");
    m1.addContact("John", 9866);
    m1.addContact("Maria", 7865);
    System.out.println("2----------------");
    m1.details();
    System.out.println("3----------------");
    m1.makeCall(9866);
    System.out.println("4----------------");
    m1.addContact("Henry", 2365);
    System.out.println("5----------------");
    m1.makeCall(7552);
    m1.makeCall(2365);
    System.out.println("6----------------");
    m1.addContact("Gomes", 4589);
    m1.addContact("Antony", 8421);
    m1.addContact("Tony", 5789);
    System.out.println("7----------------");
    m1.details();
  }
}
```

</td><td>

```
Total Contacts: 0
Contact List:
1----------------
The contact of John is added.
The contact of Maria is added.
2----------------
Total Contacts: 2
Contact List:
John:9866
Maria:7865
3----------------
Calling John . . .
4----------------
The contact of Henry is added.
5----------------
Calling 7552 . . .
Calling Henry . . .
6----------------
The contact of Gomes is added.
The contact of Antony is added.
Storage Full!!
7----------------
Total Contacts: 5
Contact List:
John:9866
Maria:7865
Henry:2365
Gomes:4589
Antony:8421
```

</td></tr>
</table>

Task 5

```java
public class B {
    public int temp = 4;
    public int sum, y, x;
    public void methodA(int m){
        int [] n = {2,5};
        int x = 0;
        y = m + this.methodB(x++,m)+(temp++);
        x = this.x + 2 + n[0];
        sum = sum + x + y;
        n[0] = sum + 2;
        System.out.println(n[0]+" " + x+ " " + sum);
    }
    public int methodB(int m, int n){
        int y = 4 + this.y + m;
        x = this.y + y + (++temp) - n;
        sum = x + y + this.sum;
        System.out.println(y+ " " + this.x + " " +sum);
        return x;
    }
}
```

```java
public class Tester11 {
    public static void main(String [] args){
        B t1 = new B();
        t1.methodA(5);
        B t2 = new B();
        t2.methodB(12, 2);
    }
}
```

Outputs		

Task 6

```
public class A{
  public int x = 3, y = 5, sum = 9;
  public int methodA(int temp, int x){
    this.x += (x++) + temp;
    if(temp % 5 == 2){
      sum += (this.y++) + temp;
    }
    else{
      sum += 3;
      if(y > 5) ++y;
    }
    System.out.println(this.x + " " + y + " " + sum);
    return this.x;
  }
  public void methodB(int y){
    int temp = (y++) + this.y;
    this.y = (++temp) + methodA(temp, y) + x;
    sum = y + (++this.x) + (temp++);
    System.out.println(x + " " + y + " " + (sum++));
  }
  public void methodC(int y){
    y = (this.x++) + sum + 3;
    System.out.println(x + " " + y + " " + sum);
  }
}
```

Driver code:

```
public class Test12{
  public static void main(String [] args) {
   A a1 = new A();
   a1.methodA(2, 4);
   a1.methodB(3);
   new A().methodC(7);
  }
}
```

Outputs		

Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

Task 1

Complete the **Bird** class so that main method produces the following **output**:

<table>
<tr><th>Test class</th><th>Output</th></tr>
<tr><td>

```
public class BirdTest{
    public static void main(String args[]) {
        Bird b1 = new Bird();
        b1.name = "Parrot";
        b1.flyUp(3);
        b1.makeNoise();
        b1.flyDown(5);
        b1.flyDown(2);
        b1.flyDown(1);
        Bird b2 = new Bird();
        b2.name = "Eagle";
        b2.flyUp(5);
        b2.flyDown(5);
        b2.makeNoise();
    }
}
```

</td><td>

```
Parrot has flown up 3 feet.
Squawk
Parrot cannot fly down 5 feet.
Parrot has flown down 2 feet.
Parrot has flown down 1 feet and
landed.
Eagle has flown up 5 feet.
Eagle has flown down 5 feet and
landed.
Squee
```

</td></tr>
</table>

Task 2

Implement the "**ChickenBurger**" class with necessary properties, so that the given output is produced for the provided driver code.
[**Note:**
1. There are four available spice levels: **Mild**, **Spicy**, **Naga** and **Extreme**. You can store these values in a String array.
2. You might need to use the *.equals()* method to compare two string values.]

<table>
<tr><th>Driver Class</th><th>Output</th></tr>
<tr><td>

```
public class BurgerMaker{
 public static void main(String [] args){
   ChickenBurger b1 = new ChickenBurger();
```

</td><td>

```
Sesame
200
Less
Not Set
----------1----------
```

</td></tr>
</table>

<pre>   System.out.println(b1.bun);
    System.out.println(b1.price);
    System.out.println(b1.sauceOption);
    System.out.println(b1.spiceLevel);
    System.out.println("----------1----------");
    System.out.println(b1.serveBurger());
    System.out.println("----------2----------");
    b1.customizeSpiceLevel("Extreme Jhaal");
    b1.customizeSpiceLevel("Spicy");
    System.out.println("----------3----------");
    System.out.println(b1.serveBurger());
    System.out.println("----------4----------");
    ChickenBurger b2 = new ChickenBurger();
    b2.bun = "Brioche";
    b2.price += 50;
    b2.sauceOption = "Regular";
    b2.customizeSpiceLevel("Naga");
    System.out.println("----------5----------");
    System.out.println(b2.serveBurger());
  }
}</pre>
	<pre>Cannot serve now. Customize Spice
	Level first.
	----------2----------
	This spice level is unavailable.
	Spice level set to Spicy.
	----------3----------
	The burger is being served:-
	Bun Type: Sesame
	Price: 200
	Sauce Option: Less
	Spice Level: Spicy
	----------4----------
	Spice level set to Naga.
	----------5----------
	The burger is being served:-
	Bun Type: Brioche
	Price: 250
	Sauce Option: Regular
	Spice Level: Naga</pre>

Task 3

Design the **MagicPotion** class so that the following Tester class generates the expected output

- Potion cannot be activated if potion strength below 5

<table>
<tr><th>Driver Code</th><th>Outputs</th></tr>
<tr><td><pre>
public class MagicTester {
  public static void main(String[] args) {
    MagicPotion potion = new MagicPotion();
    potion.setDetails("blue", 4);
    System.out.println("======================");
    potion.activate();
    potion.describe();
    System.out.println("=======================");
    potion.boost();
    potion.describe();
    System.out.println("=======================");
    potion.activate();
    potion.describe();
    System.out.println("=======================");
    System.out.println("Is potion strong? "+
potion.isStrong());
    System.out.println("=======================");
    potion.boost();
    potion.activate();
    potion.describe();
    System.out.println("=======================");
    System.out.println("Is potion strong? " +
potion.isStrong());
    System.out.println("=======================");
    potion.boost();
    System.out.println("Is potion strong? " +
potion.isStrong());
    System.out.println("========================");
    potion.weaken();
    potion.weaken();
    potion.weaken();
    potion.describe();
  }
}
</pre></td><td><pre>
======================
Potion colour: blue
Potion strength: 4
Is active: false
=======================
Potion colour: blue
Potion strength: 5
Is active: false
=======================
Potion colour: blue
Potion strength: 5
Is active: true
=======================
Is potion strong? false
=======================
Already activated
Potion colour: blue
Potion strength: 6
Is active: true
=======================
Is potion strong? false
=======================
Is potion strong? true
========================
Potion colour: blue
Potion strength: 4
Is active: false
</pre></td></tr>
</table>

Task 4

```java
public class TraceA{
  public int x =0, y = 8, sum = 0;
  public void methodA(int y, int x){
    int [] arr = {4, 7};
    this.x += x;
    arr[0] = y++;
    arr[1] += arr[1] % arr[0] * x;
    if ( y % 2 == 0){
      sum = x + methodB(arr[0]++, arr[1], y) + this.x;
    }
    else{
      sum = this.y + methodB(++arr[0], arr[1], x) + this.y;
    }
    System.out.println(x+ " " + arr[0] + " " + sum);
  }
  public int methodB(int a, int b, int x){
    sum = sum % x;
    if( a % b == x ){
      return this.y--;
    }
    this.x = a * b / x;
    y += this.x % this.y;
    System.out.println(a + " " + this.x + " " + y);
    return y;
  }
}
```

Driver code:

```java
public class TesterX{
  public static void main(String [] args){
    TraceA t1 = new TraceA();
    t1.methodA(4, 7);
    int x = t1.methodB(3,2,1);
    System.out.println(t1.y + " " + t1.sum + " " + x);
  }
}
```

Output		